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Optimized machine learning approach may improve understanding of medial tibial stress syndrome in military male personnel

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Abstract

This study investigates the applicability of optimized machine learning approach for the prediction of MTSS using anatomic and anthropometric predictors. To this end, 180 recruits were enrolled in a cross-sectional study (30 MTSS (30.36 ± 4.80 years) and 150 normal (29.70 ± 3.81 years)). Twenty-five predictors/features including demographic, anatomic, and anthropometric variables were selected as risk factors. Bayesian optimization method was used to evaluate the most applicable machine learning algorithm with tuned hyperparameters on the training data. Three experiments were performed to handle the imbalances in the data set. The validation criteria were accuracy, sensitivity, and specificity. The highest performance (even 100%) observed for the Ensemble and SVM classification models while using at least 6 and 10 most important predictors in experiments 1 and 3 respectively. In experiment 2 the best performance (accuracy = 88.89%, sensitivity = 66.67% and specificity = 95.24%) was achieved for the Naive Bayes classifier with 10 most important features. The Naive Bayes, Ensemble, and SVM methods could be the primary choices to apply machine learning approach in MTSS risk prediction. These predictive methods alongside the eight common proposed predictors might help to more accurately calculate the individual risk of developing MTSS at the point of care.

1. Introduction

Musculoskeletal injury is a serious problem in the athletic world and military organizations. The prevalence rate of these injuries is very high and unacceptable [1, 2], about ~ 25% among military males [3] and 76% among all athletes [2]. This high prevalence places a necessary demand on injured subjects and in general on societies to utilize healthcare facilities [4]. Medial tibial stress syndrome (MTSS) is one of the most common musculoskeletal injuries, the incident rate of MTSS is from 4–35% in both military and sports medicine [5, 6]. The pain associated with MTSS appears by loading impact on the lower limb during exercise, and/or marching activities [7, 8]. Scientific sources explain this syndrome as an inflammation in the medial and posterior edge of the tibia caused by repeated tension and overload impact on the lower limb [9]. Different studies mention a variety of risk factors such as navicular drop, body mass index (BMI), and age [7, 10]. However, the precise pathophysiology of this syndrome is not fully defined [10, 11] and therefore prediction of this complex and multivariable syndrome is challenging. In addition to the cost of this syndrome, the recovery time is also lengthy and extends from weeks to several months [12, 13]. Currently, MTSS is a high recurring syndrome with no reliable treatment [14].

Most of the literature focus on an ensemble of risk factors rather than a single cause for musculoskeletal injuries [15, 16]. Furthermore, the utility of the single predictor may be affected by noise and may need to reach a specific threshold to be apparent in the traditional statistical approaches. Therefore, more complex statistical methods that are used in machine learning approaches could be more appropriate and efficient to evaluate the combination of several predictors concurrently [1]. Machine learning approach includes strong analytical methods that could provide new insight into the interaction of variables. This approach has a significant potential to be used for the management of injury risks in sports medicine [17, 18].

Few studies have applied machine learning approach in MTSS risk prediction or management[18–20]. Garnock and associates [20] used 8 predictors and only utilized stepwise logistic regression for predicting the risk of injury. Other published studies have used 10 independent risk factors and reported the ranked accuracies[14, 19]. However, they failed to fully address the importance of the predictors and optimized machine learning approaches for MTSS application. Since the main predisposing factors are not yet known, there seems a necessity to do a more comprehensive study to examine the strength of the machine learning approach on the MTSS true predictors.

In the present study, we hypothesise that applying optimized machine learning approach can offer accurate results on the demographic, anatomic, and anthropometric variables as predictors of MTSS in military recruits.

2. Methods

Two hundred male personnel during combat training in the infantry brigade enrolled in this cross-sectional study. The subjects have signed a written consent form and were well informed about the project. After detailed consideration of exclusion criteria, 20 subjects have been excluded because of lower limb surgery, fracture history, and paresthesia symptoms. Therefore, the data of 180 recruits were included in the machine learning analysis. Out of 180 individual data sets, no-injury (normal) group comprised of 150 and the MTSS group of 30. The study was performed after approval by the Ethics Committee of AJA university, Tehran, Iran. Declaration of Helsinki was followed throughout the study. The MTSS was diagnosed according to the criteria put forth by Yates and White [21] including the appearance of pain following exercise lasting at least 2 hours to several days. General block diagram of the material and method section is presented in Fig. 1.

Figure 1

2.1. Data collection

The predictors were measured twice in two alternate weeks. Each predictor value was recorded as the mean of the two measurements. Predictors were of 3 categories: demographic, anatomical, and anthropometric data. Age, weight, and height were acquired by 4 skilled specialists with an average of 4 years of experience as demographic data collectors. Seca stadiometer (Germany) was used to record heights barefoot (to the nearest 0.1 cm) Weight was measured with a Seca scale (digital, Germany,) in barefoot and light clothing state (within 0.1 kg accuracy). After completion of the demographic data category, the subject was examined by a sports medicine specialist based on Moen et.al. [22] and Raissi et.al. [23]. One goniometer was used to measure angles and a tape meter was used to record all static measures (to the 0.5 cm) of the lower limb; leg length discrepancy, lower extremity length from the anterior superior iliac spine to the most prominent part of the medial malleolus. Hip range of motion (ROM) was measured when the subject was in sitting position while the knee and hip joints flexed at 90°. Hip joint has been rotated externally and internally to feel and firm [9, 22, 23]. Quadriceps (Q) angle was

calculated when the subject was in standing state with knee and hip joints fully extended and quadriceps relaxed. The arms of goniometer were placed at the lower edge of the patella and the anterior superior iliac spine, using hands as a guide, to measure the angle in degrees [23]. The intercondylar interval was measured using a gauge with 0.1 mm accuracy in subjects' standing upright while the feet were paired in one direction [23]. Intermalleolar Interval was also calculated based on Omololu et.al. study [24]. While standing, the maximum circumference of the relaxed calf was measured and recorded with a tape meter as calf girth [22]. In the same position with calf girth, proximal to the internal malleolus, the minimum circumference was recorded as ankle girth [25]. Navicular drop test (NDT) was performed when the subject was in a sitting position. Furthermore, the subject's ankle joint was in natural position and their knees flexed at 90°. A non-toxic marker has been used for marking the tuberosity of navicular bone. Next, the subjects were asked to stand without changing the position of the foot. The difference between these two positions (heights of the tuberosity of navicular bone) was considered as navicular drop in millimeters [22, 26]. Lateral tibial height, Iliospinale height (IH), and trochanteric-lateral tibial height were collected based on kinanthropometric assessments [25]. Body composition defined as the percentage of lean muscle and other body tissues was assessed with Tanita BF 350 (Japan) based on the bioelectrical impedance analysis method [27].

Data are displayed in Table 1 in both complete and abbreviated form. Values are expressed as mean ± SD.

Table 1
Twenty-five predictors were considered in the study for both MTSS (30 subjects) and normal (150 subjects) groups. Values are given as mean ± SD.

Predictor name (abbreviation)	Unit	Normal group (mean ± SD)	MTSS group (mean ± SD)
Age	year	30.36 ± 4.80	29.70 ± 3.81
Body Mass (BM)	kg	81.87 ± 14.21	79.77 ± 9.98
Stretch Stature (SS)	cm	174.15 ± 7.94	173.96 ± 6.59
Right Leg Length (RLL)	cm	87.48 ± 3.31	88.61 ± 4.70
Left Leg Length (LLL)	cm	87.87 ± 3.55	88.60 ± 4.19
Navicular Drop Test (NDT)	mm	6.19 ± 2.98	4.20 ± 3.13
Inter Condylar Interval (ICI)	cm	1.12 ± 1.51	1.31 ± 1.53
Inter Malleolar Interval (IMI)	cm	2.00 ± 3.66	1.45 ± 3.01
Q Angle (QA)	degree	14.40 ± 5.11	14.46 ± 5.51
External Rotation (ER)	degree	45.36 ± 3.64	41.46 ± 3.73
Internal Rotation (IR)	degree	38.70 ± 5.02	37.03 ± 2.73
Flexibility Right (FR)	degree	80.53 ± 11.35	81 ± 9.77
Flexibility Left (FL)	degree	81.46 ± 10.10	78.33 ± 12.75
Iliospinale Height (IH)	cm	51.15 ± 3.35	53.18 ± 2.96
Trochanteric Tibial Lateral Length (TTLL)	cm	43.02 ± 2.41	44.78 ± 1.56
Tibial Lateral Height (TLH)	cm	45.16 ± 2.54	46 ± 2.10
Bi-illiocristalis (BI)	cm	30.64 ± 1.82	30.95 ± 1.41
Femur Breadth (FB)	cm	10.08 ± 0.57	10.12 ± 0.26
Calf Girth (CG)	cm	38.13 ± 3.25	38.55 ± 3.48
Ankle Girth (AG)	cm	22.70 ± 1.42	22.71 ± 1.34
Fat Percentage (FP)	percentage	20.16 ± 5.07	19.90 ± 5.24
Fat Mass	kg	16.75 ± 6.24	15.81 ± 5.28
Fat-Free Mass (FFM)	kg	65.20 ± 8.01	64.57 ± 6.45
Fat Mass Right (FMR)	kg	2.37 ± 0.82	2.45 ± 0.91
Fat-Free Mass Right (FFMR)	kg	11.47 ± 1.46	11.11 ± 1.27

Table 1

2.2. Statistical analysis, machine learning approach

Machine learning algorithms transform the problem into an optimization problem to be solved normally. The optimization problem comprises multiple hyperparameters that are set before the training process and defines how best the model fits the data [28]. There are two main types of hyperparameters selection methods, i.e., manual and automatic search. Manual search needs highly experienced users and is hard to be applied by non-experts. To overcome the limitations of manual search, automatic methods such as grid and random searches have been proposed [29]. Random search has solved the expensive cost of exhaustive searching in grid search and proved more efficient in high dimensional space, even though it is not reliable for some complex models [29]. In addition, the problem of making the automatic tuning algorithm with high efficiency has not been fully solved in the machine learning approach [28]. Choosing the best-fitted model with tuned hyperparameters is an optimization problem with black-box objective function. Therefore, common optimization solutions like Newton or gradient descent could not be utilized.

The body of literature proposes Bayesian optimization for such kinds of problems. They suggest this method outperforms other global optimization methods [28, 30, 31]. Generally, Bayesian optimization solves a problem that function has not a closed-form. The algorithm comprises of 2 main steps (Eq. 2 and Eq. 3) introduced below where $DATA_{1:t-1} = \{x_n , y_n\}_{n=1}^{t-1}$ represents the training dataset with the *t-1* observation of unknown function.

Bayesian optimization workflow on the training dataset:
1. For *t = 1,2, …*

2. Maximizing an acquisition function (a function) over f and finding the new point asx_t

$$x_t = argmaxa\,(x|DATA_{1:t-1})\;Eq.1$$

3. Updating The Posterior Distribution

$y_t = f(x_t)$

$$DATA_{1:t} = \{DATA_{1:t-1}, (x_t, y_t)\}\;Eq.2$$

4. End For.

Our problem, i.e., classifying the MTSS group vs normal group using optimized machine learning approach falls into this category, therefore, we used Statistics and Machine Learning Toolbox™ (MATLAB and Release 2020b, The MathWorks, Inc., Natick, Massachusetts, United States) for applying automatic machine learning methods with tuned hyperparameters. The optimization algorithm is already implemented in the machine learning toolbox of MATLAB software and can be used by employing the "fitcauto" function [32–34].

As mentioned above, the Bayesian optimization goal, in general, is to find a point that objective function has extremum. In using the context of the "fitcauto" method, a point is a learner type alongside hyperparameters of the learner. The objective function is cross-validated classification error. The Bayesian optimization method in "fitcauto" internally includes a multi-TreeBagger model of the objective function (The objective function of this model differs from the Gaussian process model implemented by other machine learning toolbox functions that mainly use Bayesian optimization). In this context, an acquisition function (i.e., expected improvement) determines the next point to be evaluated. At the end of the optimization process, "fitcauto" chooses the point with the minimum objective function value, among the points evaluated during the optimization. This method evaluates seven most applicable machine learning methods (e.g., discriminant analysis classifier, ensemble classification model, k-nearest neighbor model, linear classification model, Naive Bayes classifier, support vector machine classifier (SVM), and binary decision classification tree) and automatically find the best method with tuned hyperparameters on the training data. Once the optimization process is finished the "fitcauto" returns the trained model on the entire train data set, which is expected to best classify new data. For more detail on implementing the mentioned method see Mathworks [35].

2.3. Imbalanced data sets in the machine learning approach

In our study, we had 30 MTSS subjects and 150 normal subjects, therefore the uneven sample size imposes bias in machine learning methods training. To evaluate this drawback, we assumed three different experiments. First, 30 subjects were randomly selected from 150 subjects within the normal group to equate the two groups. In the second experiment, we employed machine learning on the original imbalanced data (i.e., 30 MTSS subject's data and 150 normal subject's data). In the third case, the MTSS dataset group was randomly upsampled (i.e., generating 120 uniform distributed integer random values between 1 and 30) to 150 subjects, again equalizing the two groups but this time at 150 each. At the end, the machine learning optimization approach was employed separately on constructed predictors and response data for each experiment.

2.4 Table of predictors and response, feature importance

As mentioned in the section of data collection, the study includes 25 predictors (Table 1) and the response column (0, normal group, 1, MTSS group). We have used the filter type feature selection algorithm (e.g., feature ranking using F-tests) available in MATLAB software Statistics and Machine Learning Toolbox™. This algorithm measures the predictor importance based on the characteristics of the predictor, like feature relevance to the response or feature variance. For more information about the feature selection method see [36, 37]. We ranked features' importance as a preprocessing step and then trained the machine learning method by adding the next predictor considering the rank progressively. Twenty-five models were then generated for comparison I each experiment. For example, model 1 constructed using only predictor 1 data (i.e., the most important predictor), model 2 was constructed on both predictor 1 and predictor 2, model 3 on the first three most important predictors (i.e., 1, 2, and 3), etc. Finally, model 25 is construed using all predictors.

For each run, 85% of the dataset was used for training and the remainder (15%) for the test step. The algorithm uses the k-fold (k = 5) cross-validation method for validation of the training model. We used the "predict" command to obtain the outputs of models for the new test data. Output measures of our study to validate and estimate the effectiveness of each model were three well-known criteria i.e., sensitivity, specificity, and accuracy (Eq. 3, 4, and 5). These measures were calculated based on the confusion matrix values [38].

$$Sensitivity = \frac{TP}{TP + FN} Eq. 3$$

$$Specificity = \frac{TN}{TN + FP} Eq. 4$$

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN} Eq. 5$$

Where;

True Positive (TP): MTSS subjects correctly classified as MTSS.

True Negative (TN): Normal subjects correctly classified as Normal.

False Positive (FP): Normal subjects wrongly classified as MTSS.

False Negative (FN): MTSS subjects wrongly classified as Normal.

3. Results

As reported above, the prevalence of MTSS risk in our study was 16.16% (30 MTSS and 150 normal subjects). The table of the predictors (Table 1) contains 25 demographic, anatomical, and anthropometric variables.

We have performed 3 different experiments; before each run, the table of predictors data was constructed, and the preprocessing step calculating the importance value of each predictor was done on the table of data. Relevant results of this step (i.e., ranked predictor names and importance values) for each experiment are shown in Fig. 4.

Histogram of each predictor for both groups in 3 experiments (i.e., 30 vs 30, 30 vs 150, and 150 vs 150) is shown in Figs. 2 and 3.

Figure 2

Figure 3

Figure 4

After the preprocessing of constructed predictor data set and sorting the predictors based on the calculated importance and rank, the optimized machine learning approach methods using Bayesian optimization algorithm were implemented separately 25 times for each experiment. Results of the optimized model in each run were presented in Tables 2, 3, and 4.

Table 2
The performance of employed optimized machine learning approach in experiment 1 based on the predictor's importance and the number of predictors.

# Predictors	Accuracy (%)	Sensitivity (%)	Specificity (%)	Selected ML Model
1	44.44	33.33	66.67	KNN
2	77.78	66.67	100	Naive Bayes
3	66.67	66.67	66.67	Ensemble
4	77.78	83.33	66.67	Ensemble
5	77.78	83.33	66.67	Ensemble
6	100	100	100	Ensemble
7	100	100	100	Ensemble
8	100	100	100	Ensemble
9	100	100	100	Ensemble
10	100	100	100	Ensemble
11	100	100	100	Ensemble
12	100	100	100	Ensemble
13	100	100	100	Ensemble
14	100	100	100	Ensemble
15	100	100	100	Ensemble
16	100	100	100	Ensemble
17	100	100	100	Ensemble
18	100	100	100	Ensemble
19	100	100	100	Ensemble
20	100	100	100	Ensemble
21	100	100	100	Ensemble
22	100	100	100	Ensemble
23	100	100	100	Ensemble
24	100	100	100	Ensemble
25	100	100	100	Ensemble

Table 3
The performance of employed optimized machine learning approach in experiment 2 based on the predictor's importance and the number of predictors.

# Predictors	Accuracy (%)	Sensitivity (%)	Specificity (%)	Selected ML Model
1	77.78	0	100	Naïve Bayes
2	85.19	50	95.24	Naïve Bayes
3	85.19	33.33	100	Naive Bayes
4	81.48	33.33	95.24	Naive Bayes
5	81.48	50	90.48	Naive Bayes
6	81.48	50	90.48	Naive Bayes
7	81.48	66.67	85.71	Naive Bayes
8	85.19	66.67	90.48	Ensemble
9	85.19	66.67	90.48	Naive Bayes
10	88.89	66.67	95.24	Naive Bayes
11	88.89	66.67	95.24	Naive Bayes
12	88.89	66.67	95.24	Naïve Bayes
13	85.19	50	95.24	Naïve Bayes
14	81.48	33.33	95.24	Naïve Bayes
15	85.19	50	95.24	Ensemble
16	85.19	50	95.24	Ensemble
17	85.19	50	95.24	Ensemble
18	85.19	50	95.24	Ensemble
19	81.48	33.33	95.24	Ensemble
20	85.19	33.33	100	Ensemble
21	81.48	33.33	95.24	Ensemble
22	85.19	50	95.24	Ensemble
23	85.19	50	95.24	Ensemble
24	81.48	33.33	95.24	Ensemble
25	81.48	33.33	95.24	Ensemble

Table 4
The performance of employed optimized machine learning approach in experiment 3 based on the predictor's importance and the number of predictors.

# Predictors	Accuracy (%)	Sensitivity (%)	Specificity (%)	Selected ML Model
1	71.11	77.78	66.67	Tree
2	91.11	100	85.19	SVM
3	100	100	100	SVM
4	100	100	100	SVM
5	100	100	100	SVM
6	100	100	100	SVM
7	100	100	100	SVM
8	95.56	88.89	100	SVM
9	95.56	88.89	100	SVM
10	100	100	100	SVM
11	100	100	100	SVM
12	100	100	100	SVM
13	100	100	100	SVM
14	100	100	100	SVM
15	100	100	100	Naive Bayes
16	100	100	100	SVM
17	100	100	100	SVM
18	100	100	100	SVM
19	100	100	100	SVM
20	100	100	100	SVM
21	100	100	100	SVM
22	100	100	100	SVM
23	100	100	100	SVM
24	100	100	100	SVM
25	100	100	100	SVM

As we can see the results show the high-performance rate of using a machine learning approach for classifying the MTSS subjects from the normal group. In the first experiment the accuracy rate, sensitivity, and specificity of 100% obtained for the Ensemble classification model when using at least the 6 most important predictors (features) (see the yellow-colored columns in Table 2).

For experiment 2 the best validation results (accuracy = 88.89%, sensitivity = 66.67%, and specificity = 95.24%) were achieved for Naive Bayes classifier while the predictor data table consisting of the 10 and/or 11 or 12 most important features (see shaded rows in Table 3).

In the third experiment, the best validation parameters with the highest performance (100%) were observed while using only 3 most important predictors (see the third shaded row in Table 4). Then, we witnessed a decay in the performance when added the eighth and ninth predictors on the predictor's data table (see 8 and 9 shaded rows in Table 4). Further along, the performance was increased again while at least 10 predators were used to construct the predictor's data table (see the 10th shaded row in Table 4). The support vector machine classifier has the best performance in experiment 3.

Table 2

Table 3

Table 4

4. Discussion

The goal of the study was to evaluate the optimized machine learning approach to predict MTSS incidence out of many demographics, anatomic and anthropometric measured variables. As the results show, efficient performance, even 100%, was achieved for predicting the new cases as normal or MTSS considering underlying patterns of predictors interaction/combination with the optimized machine learning approach. The problem of how to make or use an automatic algorithm with tuned hyperparameters is a challenge [28]. Several studies report the superiority of the Bayesian optimization for solving the problem of the grid and random search and outperforming effectiveness compared to other global optimization algorithms [28–31, 33, 34, 39]. Therefore, in this investigation, the Bayesian optimization approach was used to find the best machine learning algorithms among eight important machine learning methods for predicting the risks leading to MTSS. Despite the applicability and strong qualities of machine learning in predicting the potential risks, handling imbalances in a dataset is crucial which has not been examined in relevant studies [14, 19]. To handle this drawback, we implemented 3 separate experiments and in experiment 3 the MTSS dataset was randomly upsampled to 150 subjects. Furthermore, the study results help to propose selecting the main predictors contributing most to the emergence of MTSS using the feature selection method and running the machine learning approach by progressively adding a remained new predictor 25 times. In experiment 1, the highest accuracy (100%) was achieved using at least 6 most important predictors (i.e., IH, IR, BI, FM, LLL, and TTLL) (Table 2 and Fig. 4). Results of experiment 2 emphasize the effect of the combination of 10 and/or 11 and 12 most

important predictors with the highest obtained performance (i.e., RLL, ER, BI, LLL, FB, IH, IR, FR, FM, CG, AG, and TTLL) (Table 3 and Fig. 4). The best-selected model (e.g. Naive Bayes) for our optimization machine learning approach in experiment 2 is in line with the best accuracy rate of two recent studies [14, 19]. Experiment 3 results demonstrate that in the best case using 3 most important predictors could provide the best performance and also using the combination of 10 predictors shows the highest accuracy, sensitivity, and specificity (i.e., IH, BI, IR, ER, CG, FB, RLL, TTLL, AG, FR) (Table 4, Fig. 4). The best-optimized machine learning algorithms that obtained for experiment 1 and 3 are ensemble classification model and SVM classifier respectively. While the results confirmed the hypothesis that both anatomic and anthropometric predictors have an essential contribution to estimating the risk of MTTS incidence, collection of all 25 variables is both expensive and time-consuming, especially when the target population is large. However, putting together the validation results and feature selection method (i.e., based on the predictor importance and machine learning results) of the 3 experiments and finding commonalities of the three, we propose as a trade-off the measurement/collection of the following eight predictors (IH, IR, BI, FM, TTLL, CG, FB, and ER) which we believed will provide an acceptable ensemble of variables in next study designs.

These features are not in contradiction to the relevant studies. Biomechanical dysfunction including IH was reported as a suspected variable to development of MTSS [40]. Internal hip rotation has a significant relationship with the MTSS and also hip external rotation reported as a primary risk factor for MTSS [41, 42]. Our results also confirm the importance of these two features and we emphasize acquiring IR and ER in the next studies. Further, several studies mentioned FM as the main risk factor [43]. Biomechanical studies introduced the lower limb length and other anthropometric predictors as possible risk factors in MTSS but the exact relation or significance is not well defined[44]. Results of this study propose the anthropometric parameters (e.g., BI, TTL, CG, and FB) as risk factors alongside other predictors. We claim that any single parameter may not pose a significant relation, but the association/combination of factors will enhance the prediction of MTSS. The possible justification for lower limb anthropometric predictors might be the effect of torque on these variables leading to tightness or imbalances of muscles that cause the MTSS.

There are limitations in our study; the study only considered the male personnel, as the combat brigades are all-men infantry unit. The prevalence and predictor's importance or weight might turn out to be different in female populates [45, 46]. Since the recruits were affiliated to a single military unit, results and conclusions hereby drawn are not to be generalized to other non-military populations [20, 47]. Future studies are needed to examine the replication power of this method on otherwise matched populations.

Regarding the machine learning approach, large and balanced sample sizes from multiple centers could lead to more concrete conclusions. Therefore, further investigations with a larger sample size are warranted.

5. Conclusions

Employing optimized machine learning approach method offers a more accurate risk prediction model for MTSS syndrome. It also provides us with the most important risk factors/predictors of performance in building, designing, programming, and providing enhanced physical training regiments to mitigate the incidence of musculoskeletal injuries in general and MTSS syndrome specifically. The Naive Bayes, Ensemble, and SVM could be the first choice for future studies. IH, IR, BI, FM, TTLL, CG, FB, and ER variables are important predictors to concentrate upon for future studies. These predictors and predictive methods might help military medicine officers and sports medicine clinicians to more accurately calculate the risk of MTSS at individual level at the point of care. Future investigations with large and multi-centers datasets should confirm applicability of this process and certainly determine the generalizability of the results.

Declarations

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Conflicts of interest Authors delclare no coflicts of intrest.

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Figures

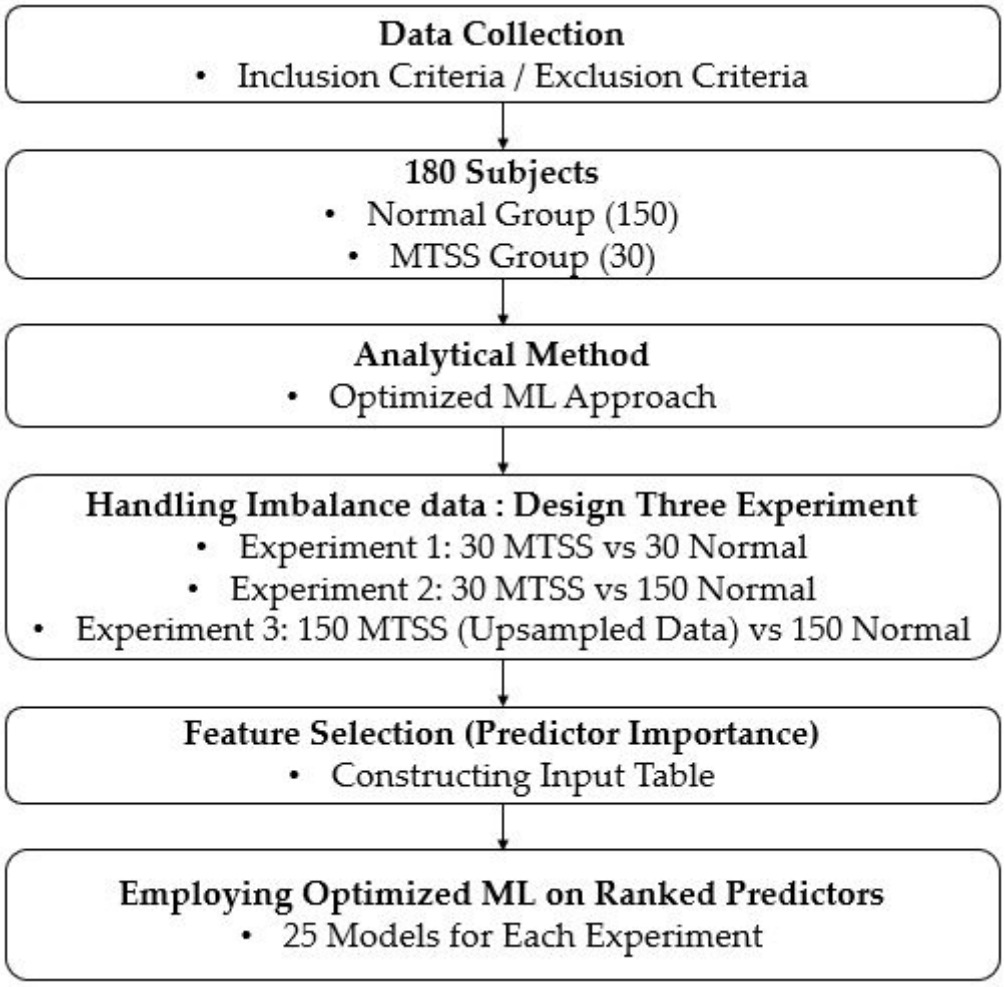


Figure 1

General block diagram of the method section.

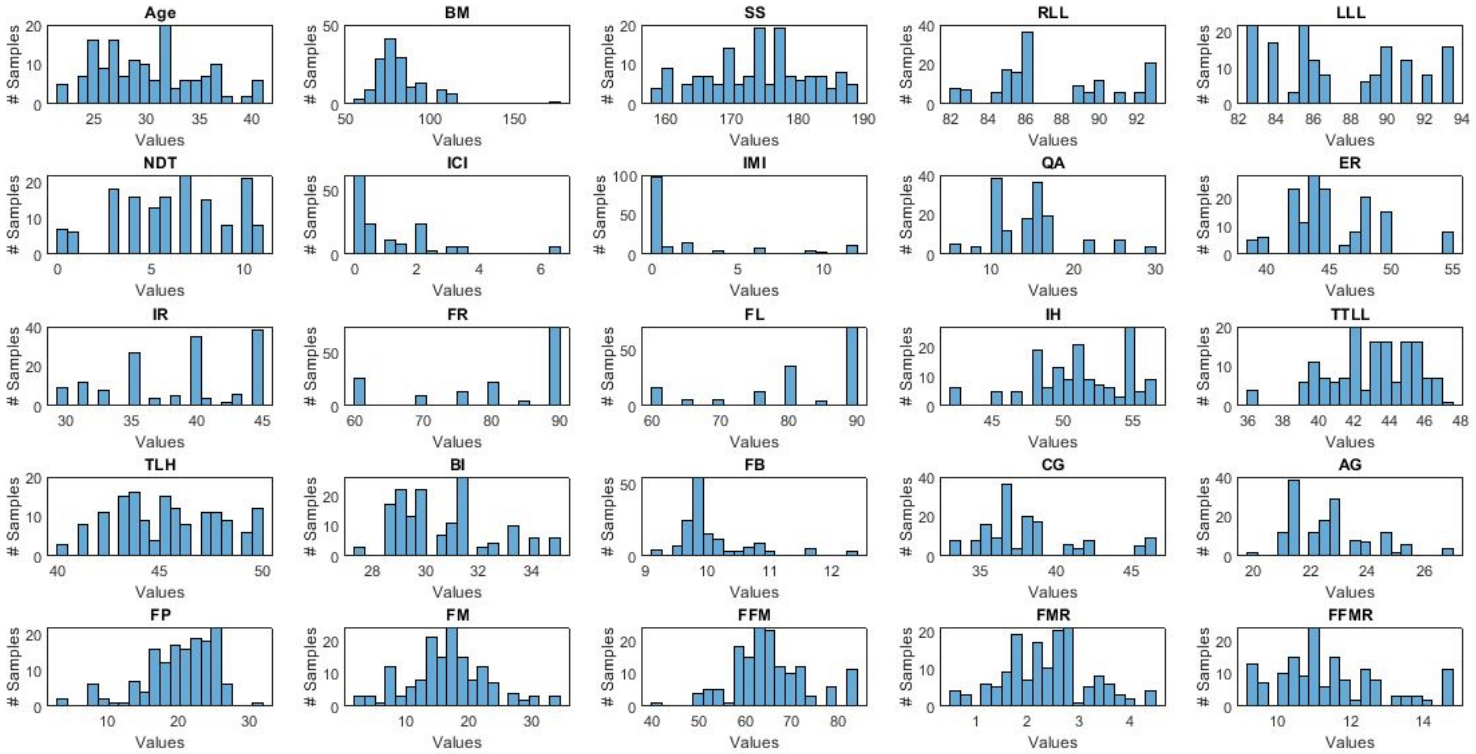


Figure 2

Histogram of the 25 predictors used for the normal group in experiments 2 and 3, Thirty subjects were randomly selected out of 150 normal subjects for experiment 1 (Note: Blue color represents normal group).

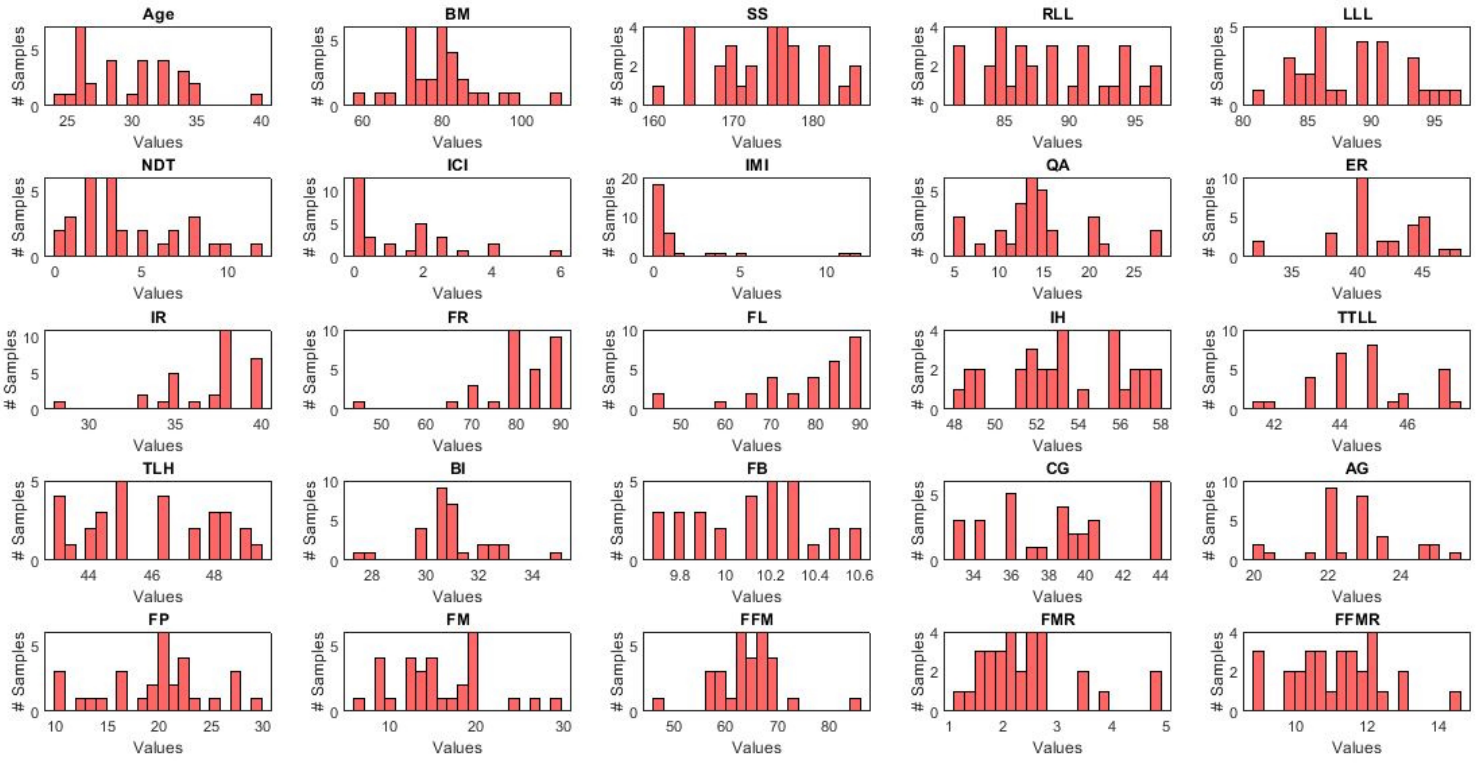


Figure 3

Histogram of the 25 predictors used for the MTSS group in experiments 1 and 2, In experiment 1 the machine learning algorithm was employed on 30 normal and 30 MTSS subjects and in experiment 2 the machine learning algorithm employed on 150 normal and 30 MTSS subjects. For experiment 3, these data were upsampled to achieve a 150 data sample size (Note: Red color represents MTSS group).

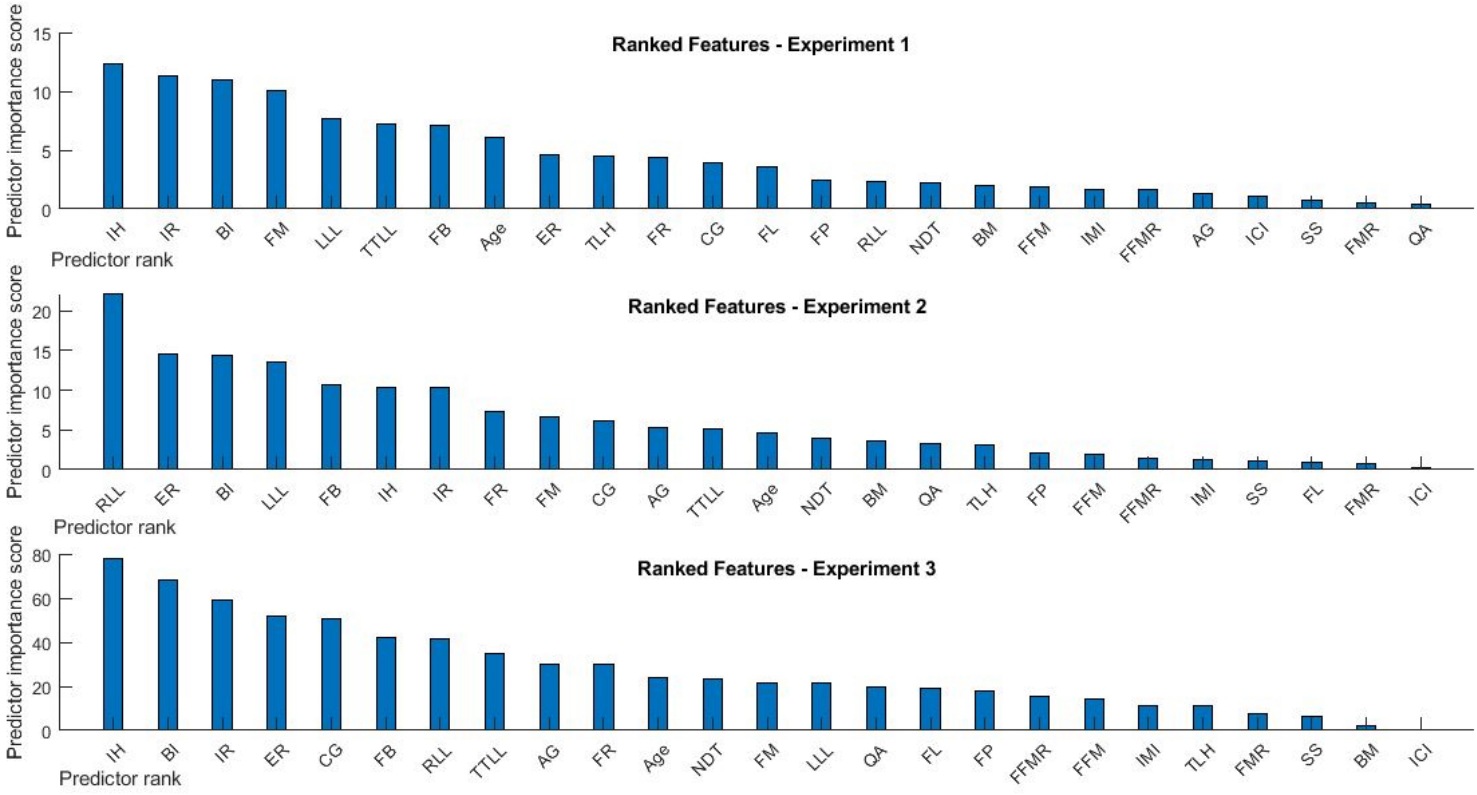


Figure 4

Predictor's importance value and rank for experiment 1 (Up), experiment 2 (Middle) and experiment 3 (Down), Examining the importance of each predictor individually using an F-test, and then rank features using the p-values of the F-test statistics calculated. The x-axis shows the predictor abbreviations/acronyms that have been sorted based on the rank and the y axis shows the importance value. The values in the y-axis are the negative logs of the p-values